

Prakash Naikade

Computer Vision Research Engineer

+49 17627767684 [✉ prakashnaikade@gmail.com](mailto:prakashnaikade@gmail.com) [🏠 Aalborg, Denmark](#)

[🌐 prakash-naikade](#) [🔄 prakashnaikade](#) [🌐 Portfolio](#)

PROFILE

I am passionate about Machine Learning, especially Computer Vision and Generative AI. I have hands-on experience from academia and industry. My research interests span in the broad areas of 3D-Reconstruction, Neural Rendering, Radiance Field, Scene Understanding, Scene Interaction, Motion Capture, AR/VR, Digital Twins, VLMs, LLMs, and generally Computer Vision, Computer Graphics, GenAI, Human Computer Interaction, XR, Deep Learning, Machine Learning, and Data Science, to solve real-world problems with impactful AI-aided solutions.

EDUCATION

PhD XRAI4AEC

Researching new paradigms at the intersection of XRAI and AEC Industry.

Aalborg University, Denmark

Aug 2025 – Aug 2028

MSc Media Computer Science

Grade: 1.6/5.0 (1.0 being the best possible score)

Saarland University, Germany

Oct 2020 – Aug 2024

Thesis: [Novel View Synthesis of Structural Color Objects Created by Laser Markings](#). (1.3)

Relevant Courses: Computer Graphics, Image Processing & Computer Vision, Neural Networks: Theory & Implementation, High-Level Computer Vision, Statistics with R, Adversarial Reinforcement Learning, Human Computer Interaction, Games & Interactive Media. [Audited]: Geometric Modeling, Machine Learning, AI, Ethics for Nerds

BEng Computer Engineering

Grade: 65% (First Class)

Pune University, India

June 2011 – May 2015

Thesis: [Secure Data Storage on Multi-Cloud Using DNA Based Cryptography](#).

Relevant Courses: Data Structures and Algorithms, Design & Analysis of Algorithms, Software Architecture, Software Engineering, Software Testing & Quality Assurance, Microprocessors & Microcontrollers

PROFESSIONAL EXPERIENCE

Junior Researcher (HiWi)

[August-Wilhelm Scheer Institute](#)

Saarbrücken, Germany

Sept 2023 – Dec 2024

- Worked on several applied research projects, including MediHopps, iperMö, FläKI and VuLCAn.
- Implemented advanced deep learning methods for human action recognition (HAR) and body pose estimation (HPE), and delivered detailed performance evaluations of these models, along with a trained HAR model (ST-GCN++) for custom rehabilitation exercise data captured in the lab.
- Contributed significantly to the feature extraction, generation, and visualization of furniture functionalities in the Python codebase for the iperMö project, developing an AR application to turn individual furniture wishes into reality.
- Systematic Literature Research and Reviews, Project Proposals and Scientific Literature Writing.
- Generally worked on computer vision, computer graphics, machine/deep learning, data science and some XR tasks.

Research Assistant

[AIDAM, Max Planck Institute for Informatics](#) Advisor: [Dr Vahid Babaei](#)

Saarbrücken, Germany

July 2023 – Aug 2024

- Worked on Radiance Field methods for Novel View Synthesis of structural color objects created by laser markings.
- Benchmarked SOTA radiance methods for synthetic scene involving Structural Color Object created in Blender.
- Developed capture setup to capture highly reflective and shiny structural color paintings on metal substrates.
- Improved the scene optimization using geometric prior and Anisotropy Regularizer in 3D Gaussian-Splatting method.
- Presented comprehensive experiments to demonstrate methods for simulating structural color objects before printing them using only captured images of laser-printed primaries.
- Facilitated interactive visualization of view-dependent structural color objects in web viewer.

Computer Vision Intern

[BASF-Coatings GmbH](#)

Münster, Germany

March 2023 – May 2023

- Developed dataset for adhesive test and corrosion detection on images of test panels of metal substrates.
- Developed framework and trained YOLOv8 model for adhesive tests' detection and UNet for corrosion detection using created dataset for automation project.

Computer Vision Intern

[Fenris GmbH](#)

Aachen, Germany

May 2022 – Sept 2023

- Contributed to the development of Motion2Coach project, a markerless motion capture solution using single and multiple cameras for athlete motion tracking and analysis.
- Conducted a comprehensive literature research and review focused on deep learning approaches for human pose estimation, and benchmarked SOTA approaches for domain-specific video data captured in the lab.
- Worked on various tasks such as camera calibration, deep learning based human pose estimation & golf sequence detection, comparing two pose sequences and visualization of results in Blender and Unity.

During the preparation of this [exam](#), I gained undergraduate level knowledge of Anthropology, Polity, Governance, Indian Constitution, Social Justice, International Relations, Economics (Macro), Indian & World Geography, Indian & World History, Indian Culture & Society, Environment, and Ethics. (Overall pass percentage of candidates $\approx 0.1\%$)

SKILLS

- **Programming:** Python, C#, C++, R, SQL, Matlab
- **Frameworks:** PyTorch, TensorFlow, NumPy, Pandas, sklearn, skimage, OpenCV, Open3D, Matplotlib, HuggingFace
- **Tools:** Conda, Jupyter Notebook, Git, Unity, Blender, Metashape, Colmap, Meshlab, Docker, Slurm/HPC, DevOps
- **OS:** Linux, Windows, Shell/Dos Scripting
- **Concepts:** Regression, k-NN, k-Means Clustering, PCA, SVM, Neural Networks, CNN, RNN, LSTM, Transformers, ViT, CLIP, Autoencoders, VAE, GAN, Diffusion Models, LLM, VLM, NLP, GPT, Prompt Engineering, LangChain, 2D/3D Image Processing, Object Detection, Classification, Localization, Segmentation, NeRF, 3DGS, 3D Reconstruction, PCD, Scene Understanding, Scene Interaction, HCI, XR, Reinforcement Learning, Database Design, ETL, CI/CD

PROJECTS

Learn-LLMs

GenAI, Information Retrieval

Getting a hands-on experience of using different LLM models and tools, to understand the finetuning, data preparation, evaluation & other techniques related to LLMs such as RAG and AI Agents.

Roman Villa Nennig Bot - Your virtual guide to Roman Villa Nennig

NLP

This chatbot helps the user throughout their journey of visiting a museum of the Roman Villa Nennig, developed using Google Cloud, Dialogflow Essentials and Telegram.

3D-AI

3D Computer Vision, GenAI

In this project I implement basic Deep Learning, 3D Computer Vision (3D CV), Graphics and other techniques for 3D AI.

Diffusion Models

Computer Vision, GenAI

This Project is a basic implementation of Diffusion Model to understand how diffusion works.

Human Action Recognition (HAR)

Computer Vision

Investigating the performance of different deep learning models and their ensembles used for HAR in still images.

Image Segmentation on PASCAL VOC and Cityscapes Datasets

Computer Vision

Training and Evaluation of CNNs like UNet, RU-Net and R2U-Net for Image Segmentation.

Ludwig Palette - an AR painting game

AR/VR

App developed in Unity and C# allows visitors of Ludwigskirche to explore its architecture by painting on its surfaces and understand the intricacies of sculptures inside the church.

Mini-RayTracer

Computer Graphics

Simple ray tracing engine developed in C++.

Synthetic Dataset

Computer Graphics

Generate simple 3D rendered datasets in Blender and Unity.

Easy Flappy Bird

Game Development

An simple implementation of Flappy Bird game using Unity and C#.

Movement Pattern Explorer

R, Data Visualization

A R Shiny MoveApp that enables users to interactively explore and visualize animal movement data from Movebank.

PUBLICATIONS

- HCII, May 2025. **Exploring the Potential of Virtual Reality and Artificial Intelligence in Physical Rehabilitation: Benefits, Challenges, and a Tailored Application Design.** N Dahlem, **P Naikade**, L Steffny, K Gisa, K Theilmann, D Werth.
- IJAERD, March 2015. **Secure Data Storage on Multi-Cloud Using DNA Based Cryptography.** D Zingade, S Dhuri, **P Naikade**, N Gade, A Teke.

CERTIFICATIONS

- **Kaggle:** Python, ML, Pandas, Feature Engineering, Data Visualization, Data Cleaning, SQL, Reinforcement Learning & Game AI, Time Series • **Udacity:** C++, AWS ML Foundations
- **Coursera:** Mathematics for Machine Learning and Data Science, Structuring ML Project, Neural Network and Deep Learning, Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
- **Udemy:** Foundations of MR, XR, VR Development on Quest headsets with Meta's Presence Platform and Unity
- **DataCamp:** Intermediate R, Data in R • **Memgraph:** Graph Analytics

LANGUAGES

English (Fluent), **Hindi** (Fluent), **Marathi** (Native), **German** (Elementary)

HOBBIES

Biking, Running, Hiking, Movies, Music